

Zewei Zhang

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EDUCATION

PhD , University of Iowa, Iowa City, IA <u>Major</u> : Cell and Developmental Biology (GPA: 3.9/4.0) <u>Advisor</u> : Dr. Thomas Rutkowski, Department of Anatomy and Cell Biology	08/2022 – 05/2027 (expected)
MS , Zhengzhou University Shanghai Jiao Tong University School of Medicine, China <u>Major</u> : Plastic Surgery <u>Advisor</u> : Dr. Guangshuai Li Dr. Tao Zan	08/2019 – 05/2022
MD , Zhengzhou University, China <u>Major</u> : Clinical Medicine	08/2014 – 05/2019

RESEARCH FOCUSES

- Protein folding; post-translational modifications; endoplasmic reticulum stress; stem cells; liver disease
- Epigenetic gene regulation; single-cell transcriptomics; inflammation; skin disease; cancer; clinical trials

RESEARCH SKILLS

- *In vitro* experiments (cell proliferation & migration, qPCR & WB, ELISA, flow cytometry, IHC/IF staining, and confocal microscopy)
- *In vivo* experiments (mice)
- *In silico* bioinformatics (bulk & single-cell RNA sequencing, GO, KEGG, PPI, R, and Python)

RESEARCH EXPERIENCES

Graduate Research Assistant Department of Anatomy and Cell Biology, University of Iowa, Iowa City, IA	08/2022 – 05/2027 (expected)
• Designed and executed 150+ <i>in vitro</i> and <i>in vivo</i> experiments using cellular and molecular biology techniques (qPCR, Western Blot, Flow Cytometry) to track post-translational dynamics and proteostatic stress in stem cells and liver injury, leading to 2 peer-reviewed manuscripts and 2 conference abstracts. • Synthesized experimental and analytical results into 100+ slide decks to translate complex scientific data into actionable insights and research strategies, effectively communicating findings to 10+ multidisciplinary teams of biologists and data scientists. • Managed 5+ concurrent preclinical projects and coordinated 50+ strategy meetings annually in a collaborative research environment, investigating how specific genetic or pharmacologic perturbations affect cellular or molecular interactions.	
Graduate Research Assistant Zhengzhou University Shanghai Jiao Tong University School of Medicine, China	04/2019 – 05/2022
• Integrated transcriptomic (bulk & single-cell) and epigenetic (non-coding RNAs and post-translational modifications) profiling in 200+ skin and cancer patient tissues, resulting in 8+ peer-reviewed papers. • Developed and executed robust bioinformatics workflows (Python and R on HPC) to analyze multi-omics datasets, including bulk or single-cell RNA sequencing of cellular populations with 20,000+ cells and cellular/molecular pathway analysis with 100+ candidate markers. • Designed and performed 100+ <i>in vitro</i> and <i>in vivo</i> experiments through histologic technology (H&E/ICC/IHC/IF staining), biomicroscopic imaging (fluorescent and confocal microscopy), and flow cytometry to demonstrate functional impacts of genetic or pharmacologic perturbations.	

WORK EXPERIENCES

Associate Editor05/2022 – Present

Springer Nature | Oxford University Press, Remote

- Reviewed 150+ preclinical, clinical, and translational manuscripts, including high-throughput sequencing, single-cell and spatial proteomics, liquid chromatography and mass spectrometry, protein interactions, and post-translational modifications across multiple disease states (skin, liver, neuroscience, and cancer).
- Organized 10+ challenging publication projects every month by collaborating with 50+ cross-functional partners, including scientists in the global and US biotechnology industry.

HONORS & AWARDS

1. Graduate Leadership Award, by the Department of Anatomy and Cell Biology at the University of Iowa, \$250 (05/2026)

2. Trainee Travel Award for "Endoplasmic Reticulum Conference", by Federation of American Societies for Experimental Biology, \$500 (06/2025)

SERVICES & LEADERSHIPS

Professional Services

Journal Editor

Journal of Translational Medicine09/2024 – Present

Clinical and Experimental Dermatology10/2022 – Present

Manuscript Reviewer (Total: 172 peer reviews for 37 journals)

iMeta, Nucleic Acids Research, Frontiers in Immunology, Journal of Translational Medicine, Rheumatology, Frontiers in Cell and Developmental Biology, Journal of Medical Internet Research, Frontiers in Oncology, Gene, Journal of Biological Chemistry, BMC Genomics, Oxidative Medicine and Cellular Longevity, Cellular Signalling, Frontiers in Genetics, Epigenomics, Aesthetic Surgery Journal, STAR Protocols, Frontiers in Molecular Biosciences, PeerJ, Plos One, BioMed Research International, Stem Cells International, Clinical and Experimental Dermatology, Dermatologic Therapy, JMIR Medical Informatics, JCO Clinical Cancer Informatics, Cancer Biomarkers, Clinical, Cosmetic and Investigational Dermatology, BMC Cardiovascular Disorders, Heliyon, JMIR Research Protocols, Interactive Journal of Medical Research, Health Science Reports, JMIR Medical Education, International Journal of Data Mining and Bioinformatics, JMIR Dermatology, Brain-X

Institutional Services

Graduate Student Senator08/2025 – Present

Department of Anatomy and Cell Biology Student Government | University of Iowa

- Led the department's first-ever PhD career exploration workshop, informing 10+ graduate students about diverse career pathways in academia and industry.
- Participated in Graduate Student Senate meetings every month, disseminating monthly updates to 20+ graduate students for communication and engagement.

Distribution Manager08/2025 – Present

Editorial Committee, Graduate & Professional Student Government, University of Iowa

- Coordinated the launch of the university's first-ever, graduate student-led periodical, elevating scholarly and community voices from 15+ graduate students and faculty members.
- Managed and executed the distribution of 1,200+ print copies to 5+ university libraries and 50+ academic programs, enhancing engagement and campus-wide visibility.

TEACHING EXPERIENCES

Teaching Assistant

08/2025 – 12/2025

University of Iowa, Iowa City, IA

Course: Mechanisms of Cellular Organization

Students: 15 PhD students

PUBLICATIONS

(Total: 11)

Peer-reviewed Journal Articles

1. Huang, X., Zhao, Y., Liu, D., Gu, S., Liu, Y., Khoong, Y., Luo, S., **Zhang, Z.**, ..., & Zan, T. (2023). ALKBH5-mediated m6A demethylation fuels cutaneous wound re-epithelialization by enhancing PELI2 mRNA stability. [Inflammation and regeneration](#). (IF: 6.6)
 - Post-translational modifications; Multi-omics; Protein interactions; Dermatology.
2. Xia, W., Liu, Y., Jiang, X., Li, M., Zheng, S., **Zhang, Z.**, ..., & Zan, T. (2023). Lean adipose tissue macrophage derived exosome confers immunoregulation to improve wound healing in diabetes. [Journal of Nanobiotechnology](#). (IF: 9.4)
 - Epigenetic gene regulation; Multi-omics; Dermatology; Immunology.
2. Zhao, Y.[†], Huang, X.[†], **Zhang, Z.**[†], Li, H., & Zan, T. (2022). The Long Noncoding Transcript HNSCAT1 Activates KRT80 and Triggers Therapeutic Efficacy in Head and Neck Squamous Cell Carcinoma. [Oxidative Medicine and Cellular Longevity](#). (IF: 7.3)
 - Epigenetic gene regulation; Post-translational modifications; Cancer.
3. **Zhang, Z.**, Huang, X., ..., & Li, G. (2021). Identification and functional analysis of a three-miRNA ceRNA network in hypertrophic scars. [Journal of Translational Medicine](#). (IF: 8.4)
 - Epigenetic gene regulation; Dermatology (skin cell biology); Bioinformatics.
4. Huang, X., Gu, S., Liu, C., Zhang, L., **Zhang, Z.**, ..., & Zan, T. (2021). CD39+ Fibroblasts Enhance Myofibroblast Activation by Promoting IL-11 Secretion in Hypertrophic Scars. [Journal of Investigative Dermatology](#). (IF: 7.5)
 - Gene regulation; Dermatology (molecular & cell biology); Immunology.
5. Gu, S., Huang, X., Xu, X., Liu, Y., Khoong, Y., **Zhang, Z.**, ..., & Zan, T. (2021). Inhibition of CUB and sushi multiple domains 1 (CSMD1) expression by miRNA-190a-3p enhances hypertrophic scar-derived fibroblast migration in vitro. [BMC Genomics](#). (IF: 4.5)
 - Epigenetic gene regulation; Dermatology (skin cell biology); Bioinformatics.
6. Gu, L.[†], Wang, P.[†], Du, Q.[†], **Zhang, Z.**[†], An, Y., Li, G., & Liu, L. (2021). Thirty Years Later: What Has Craniofacial Distraction Osteogenesis Surgery Replaced? [Plastic and Reconstructive Surgery](#). (IF: 5.1)
 - Plastic surgery (tissue regeneration).
7. Zhao, Y.[†], Huang, X.[†], **Zhang, Z.**[†], ..., & Li, Q. (2020). USP15 Enhances Re-epithelialization Through Deubiquitinating EIF4A1 During Cutaneous Wound Repair. [Frontiers in Cell and Developmental Biology](#). (IF: 6.0)
 - Protein interactions; Post-translational modifications; Dermatology; Bioinformatics.
8. Shi, K., Zhu, X., Liu, Z., Sun, N., Gu, L., Wei, Y., Cheng, X., **Zhang, Z.**, ..., & Liu, L. (2020). Clinical characteristics of malignant melanoma in central China and predictors of metastasis. [Oncology Letters](#). (IF: 3.1)
 - Dermatology; Cancer (melanoma biomarkers).

[†] Equal contributions

Manuscripts in Progress and Under Review

1. Huang, X., ..., **Zhang, Z.**[‡], Wang, M.[†], & Zan, T.[‡]. CTHRC1+ Fibroblasts Orchestrate the Fibrogenic Niche in Pathological Skin Scarring through a Self-reinforcing TGF- β Feedback Loop. *Under Review at International Journal of Biological Sciences*.
 - Single-cell transcriptomics; Post-translational modifications; Protein Interactions; Dermatology.
2. Velarde, T. F., Liu, K., **Zhang, Z.**, ..., & Rutkowski, D. T. CHOP promotes the transition to chronic integrated stress response signaling with suppression of hepatocyte identity. *Under Review at eLife*.
 - Protein folding; Post-translational Modifications; Endoplasmic Reticulum Stress; Liver.
3. **Zhang, Z.**, ..., & Rutkowski, D. T. Cell Differentiation Sensitizes Endodermal and Hepatic Cells to Endoplasmic Reticulum Stress. *In Progress*.
 - Post-translational Modifications; Unfolded Protein Response; Endoplasmic Reticulum Stress; Stem Cells; Liver.

CONFERENCE PRESENTATIONS

(Total: 2)

1. **Zhang, Z.**, ..., & Rutkowski, D. T. (2025). A PKA-YAP axis links hepatocyte differentiation state to ER stress sensitivity. Poster presentation at the Endoplasmic Reticulum Conference by the Federation of American Societies for Experimental Biology, Melbourne, Florida, June 2025.
 - Protein Interactions; Post-translational modifications; Endoplasmic Reticulum Stress; Liver.
2. Huang, X.[†], **Zhang, Z.**[†], ..., Amendt, B. A., & Zan, T. (2023). Integrated single-cell transcriptomic analysis discloses homogenous fibroblast landscape between human scars. Poster presentation at The Conceptual Power of Single Cell Biology by Cell Symposia, San Diego, California, August 2023.
 - Single-cell transcriptomics; Post-translational modifications; Protein Interactions; Dermatology.

[‡] Co-Correspondence

[†] Equal contributions